https://github.com/BryCor/CIS-344-GROUP-PROJECT

# PET ADOPTION SYSTEM

***Web-Based Application Project Report***

**Course Name:** CIS 344

**Instructor:** Prof. Yanilda Peralta Ramos


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# 1. Project Plan

## Project Scope:

The Pet Adoption System is an internet-based software that will be employed in the process of pairing adopters and pets. Administrators are able to control records of pets, adopters, and adoption applications.

## Project Objectives:

- Develop a pet and adopter management system that is user friendly.
- Save the information of the stores in MySQL database.
- CRUD operations of pets, adopters, and applications.
- Have a dashboard that has the required statistics and fast navigation.

## Initial Design Plans:

- Frontend (HTML/CSS), Backend (PHP), Database (MySQL).
- The relationship between pets, adopters, and applications was created using tables that were linked to a foreign key.
- Simple navigation on the Dashboard(index.php), Pets, Adopters, and Applications pages.

# 2. Technical Details

## Development Process (Step-by-Step):

1. Model the MySQL database, which has pets, adopters, and applications tables.
2. Build the back-end based on PHP to be able to communicate with databases.
3. design, style, HTML/CSS.
4. CRUDs of Frontend/Backend Integration Test.
5. Full sample data and characteristics of tests.

## Backend:

- The connection to the database is db.php.
- CRUD is performed in the PHP scripts in index.php, pets.php, adopters.php, applications.php and dp.php.
- There is select, insert, update, delete and join in queries.

## Database (MySQL): Michael Lartey

The database was structured using a relational schema with three main tables:

- Adopter
- Application
- Pets

**Relationships:**

- applications. adopter_id → adopters. adopter_id
- applications. pet_id → pets. Pet_id

```sql
-- ====================
-- JOIN QUERY
-- ====================

SELECT
    a.application_id,
    ad.Name,
    p.name AS pet_name,
    a.status
FROM applications a
JOIN adopters ad
    ON a.adopter_id = ad.adopter_id
JOIN pets p
    ON a.pet_id = p.pet_id;
    SELECT
    ad.Name,
    p.name,
    a.application_date
FROM applications a
JOIN adopters ad ON a.adopter_id = ad.adopter_id
JOIN pets p ON a.pet_id = p.pet_id;
```

**Use of JOINs:**

JOIN queries were used to combine data from multiple tables instead of storing redundant information. This allows the application to dynamically retrieve meaningful data, such as which adopter applied for a specific pet.

Using JOINs improves:

- Data organization
- Query efficiency
- Overall database design quality

### Frontend/Backend (HTML/PHP): Bryan Cortes

- Dashboard displays the number of pets, adopters, and applications.
- Click the Applications management page and combine both pets and adopters' information.
- Real-time validation on add/ update record forms and approving applications.

## 3. Code Snippets

**Database Connection (db.php):**

```php
db.php
<?php
// connect to database
$conn = mysqli_connect("localhost", "root", "", "pet_adoption_db");
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
?>
```

**Dashboard (index.php):**

```html
<!-- Sidebar -->
<aside class="sidebar">
    <div class="sidebar-brand">
        <div class="brand-icon">🐾</div>
        <div>
            <div class="brand-text">PetAdopt</div>
            <div class="brand-sub">Adoption System</div>
        </div>
    </div>
    <nav class="sidebar-nav">
        <a href="index.php" class="active"><span class="nav-icon">🏠</span> Home</a>
        <a href="pets.php"><span class="nav-icon">🐕</span> Pets</a>
        <a href="adopters.php"><span class="nav-icon">👤</span> Adopters</a>
        <a href="applications.php"><span class="nav-icon">📋</span> Applications</a>
    </nav>
    <div class="sidebar-footer">
        <a href="#">🔒 Logout</a>
    </div>
</aside>
```

This pet adoption application is split into a few simple PHP pages. index.php is the home page that shows links to Pets, Adopters, and Applications. db.php connects the site to the MySQL database, and each page includes it to run queries.

**Pets Management (pets.php):**

```php
pets.php
<?php
include 'db.php';

if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $name = $_POST['name'];
    $species = $_POST['species'];
    $breed = $_POST['breed'];
    $age = $_POST['age'];
    $gender = $_POST['gender'];

    $sql = "INSERT INTO pets (name, species, breed, age, Gender) VALUES ('$name', '$species',
    '$breed', '$age', '$gender')";
    mysqli_query($conn, $sql);
    header("Location: pets.php");
    exit;
}

$sql = "SELECT * FROM pets";
$result = mysqli_query($conn, $sql);
$pet_count = mysqli_num_rows($result);
?>
```

pets.php lets a user add a new pet using a form, saves it in the pets table, and then displays all pets in a table. adopters.php works the same way, but for adopters, saving name and email into the adopters table and lists all adopters.

**Adopters Management (adopters.php):**

```php
adopters.php
<?php
include 'db.php';

if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $name = $_POST['name'];
    $email = $_POST['email'];

    $sql = "INSERT INTO adopters (Name, Email) VALUES ('$name', '$email')";
    mysqli_query($conn, $sql);
    header("Location: adopters.php");
    exit;
}

$result = mysqli_query($conn, "SELECT * FROM adopters");
$adopter_count = mysqli_num_rows($result);
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="Manage adopters in the Pet Adoption System.">
    <title>Adopters – Pet Adoption System</title>
    <link rel="stylesheet" href="styles.css">
</head>
```

## Applications Management (applications.php):

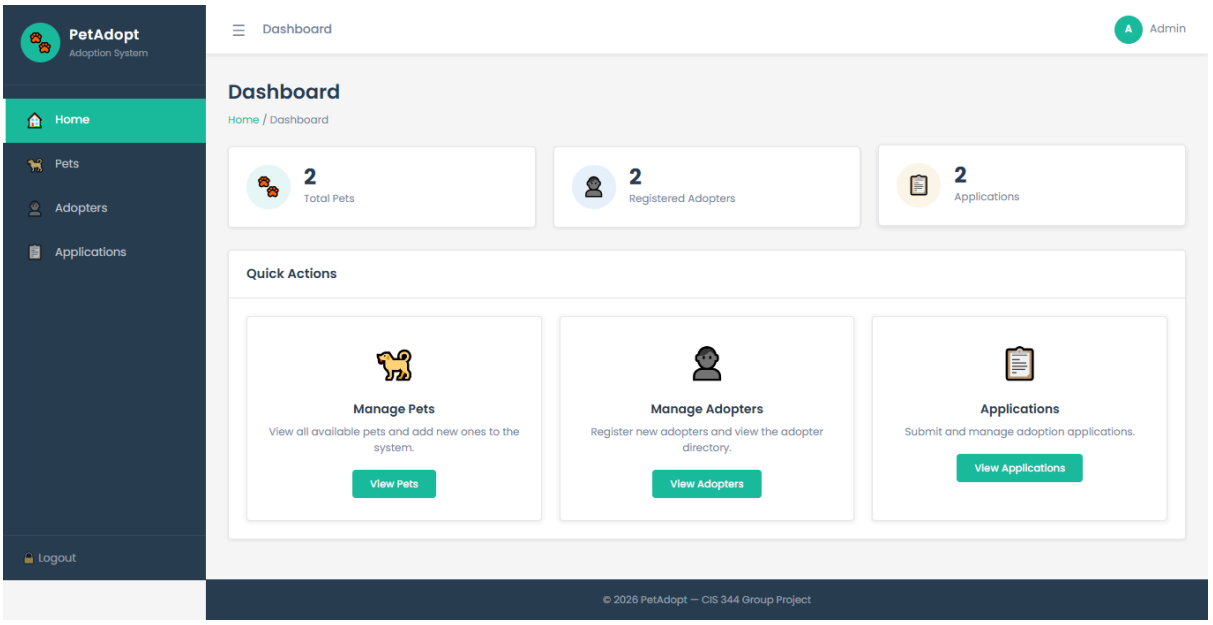
```php
<?php
include 'db.php';

if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    if (isset($_POST['action'])) {
        $app_id = $_POST['application_id'];
        if ($_POST['action'] == 'approve') {
            mysqli_query($conn, "UPDATE applications SET status='Approved' WHERE application_id=$app_id");
        } elseif ($_POST['action'] == 'deny') {
            mysqli_query($conn, "UPDATE applications SET status='Denied' WHERE application_id=$app_id");
        }
        header("Location: applications.php");
    } else {
        $adopter_id = $_POST['adopter_id'];
        $pet_id = $_POST['pet_id'];
        $sql = "INSERT INTO applications (adopter_id, pet_id) VALUES ($adopter_id, $pet_id)";
        mysqli_query($conn, $sql);
        header("Location: applications.php");
    }
}

$adopters = mysqli_query($conn, "SELECT adopter_id, Name as full_name FROM adopters");
$pets = mysqli_query($conn, "SELECT pet_id, name FROM pets");
$apps = mysqli_query($conn, "SELECT a.application_id, b.Name as full_name, p.name, p.species, a.application_date, a.status FROM applications a JOIN ad
?>
<!DOCTYPE html>
<html>
<head>
    <title>Applications</title>
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <h1>Applications</h1>
    <a href="index.php">Home</a> | <a href="pets.php">Pets</a> | <a href="adopters.php">Adopters</a>

    <h2>Submit Application</h2>
    <form method="post">
        Adopter: <select name="adopter_id" required>
            <?php while ($row = mysqli_fetch_assoc($adopters)) { ?>
                <option value="<?php echo $row['adopter_id']; ?>"><?php echo $row['full_name']; ?></option>
            <?php } ?>
        </select><br>
        Pet: <select name="pet_id" required>
            <?php while ($row = mysqli_fetch_assoc($pets)) { ?>
                <option value="<?php echo $row['pet_id']; ?>"><?php echo $row['name']; ?></option>
            <?php } ?>
        </select><br>
        <input type="submit" value="Submit">
    </form>

    <h2>Applications List</h2>
    <table class="table">
        <tr>
            <th>ID</th>
            <th>Adopter</th>
            <th>Pet</th>
            <th>Species</th>
            <th>Date</th>
            <th>Status</th>
            <th>Actions</th>
        </tr>
        <?php while ($row = mysqli_fetch_assoc($apps)) { ?>
        <tr>
            <td><?php echo $row['application_id']; ?></td>
            <td><?php echo $row['full_name']; ?></td>
            <td><?php echo $row['name']; ?></td>
            <td><?php echo $row['species']; ?></td>
            <td><?php echo $row['application_date']; ?></td>
            <td><?php echo $row['status']; ?></td>
            <td>
                <?php if ($row['status'] == 'Pending') { ?>
                    <form method="post" style="display:inline;">
                        <input type="hidden" name="application_id" value="<?php echo $row['application_id']; ?>">
                        <input type="submit" name="action" value="approve">
                        <input type="submit" name="action" value="deny">
                    </form>
                <?php } ?>
            </td>
        </tr>
        <?php } ?>
    </table>
</body>
</html>
```

applications.php is where adoption applications are created and managed. A user selects an adopter and a pet from dropdown menus, then submits an application, which is inserted into the applications table. The page also shows an applications list using SQL joins, so it can

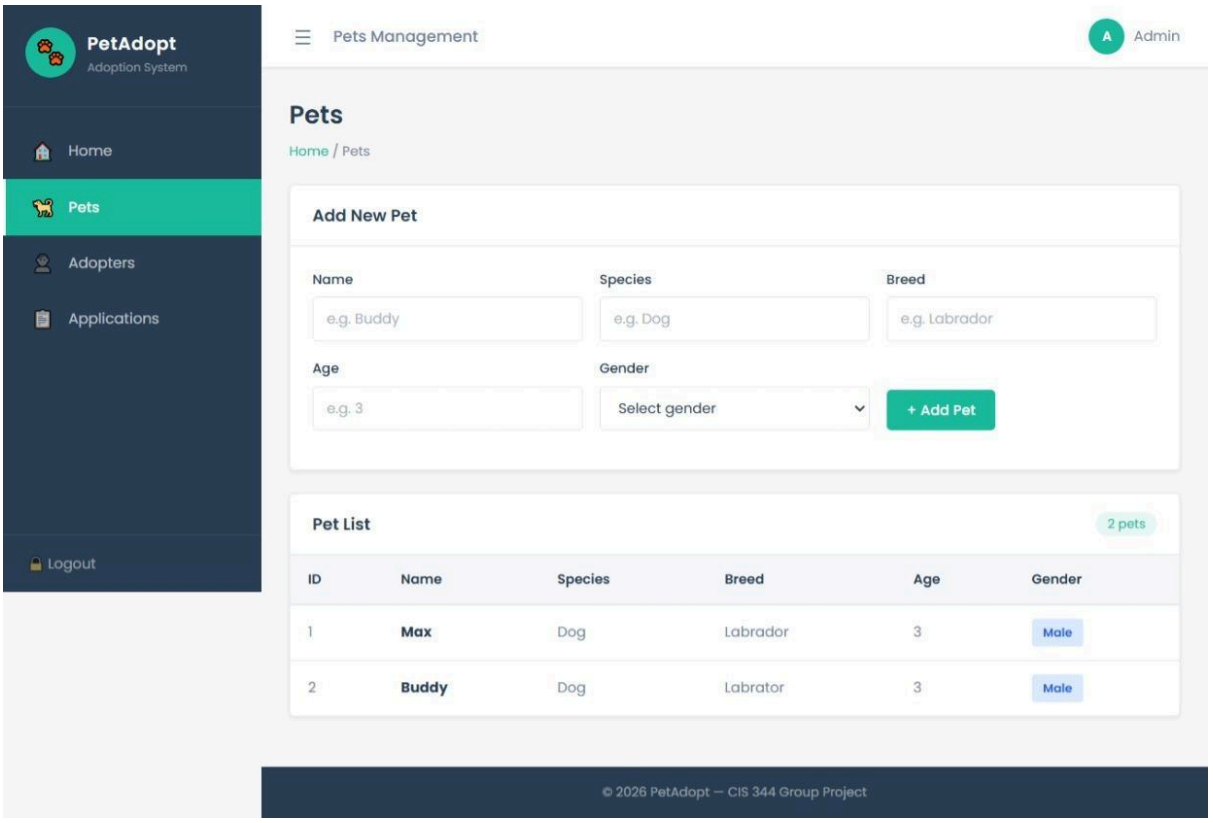
display adopter information and pet details together. If an application is still pending, the page provides Approve and Deny buttons that update the status in the database.

# 4. Application Snips/Styling(CSS): Tahsan Mynur

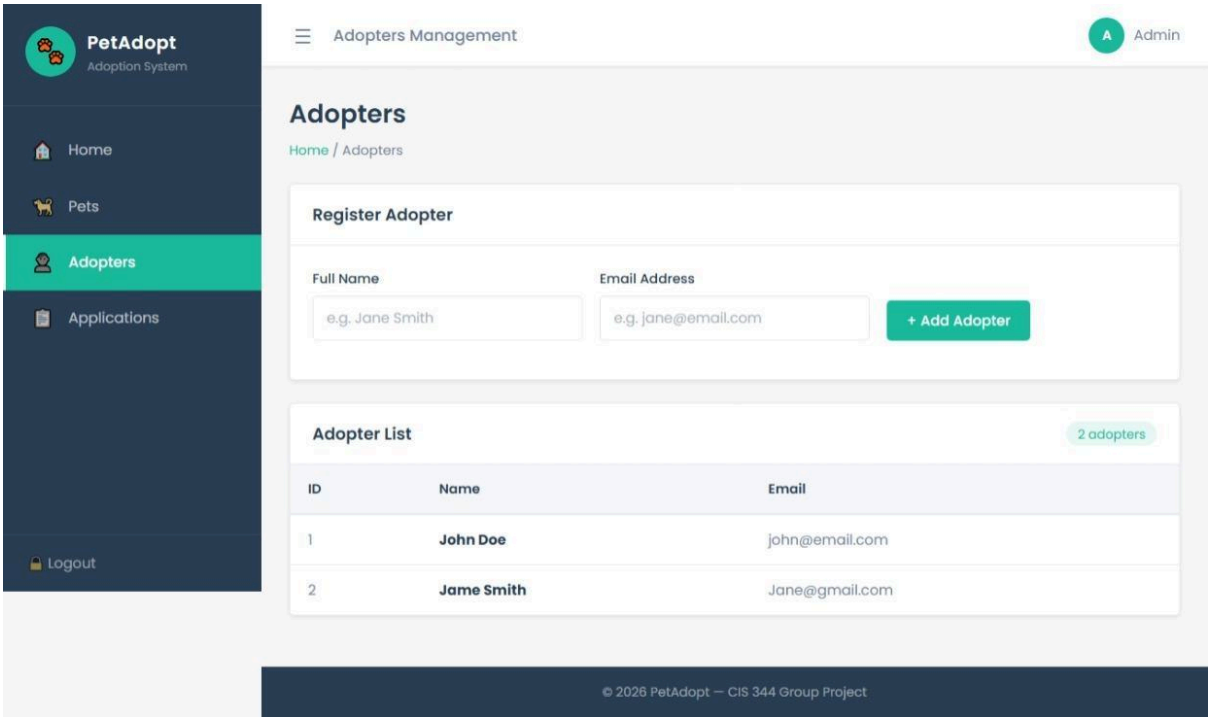
## Dashboard Screenshot:



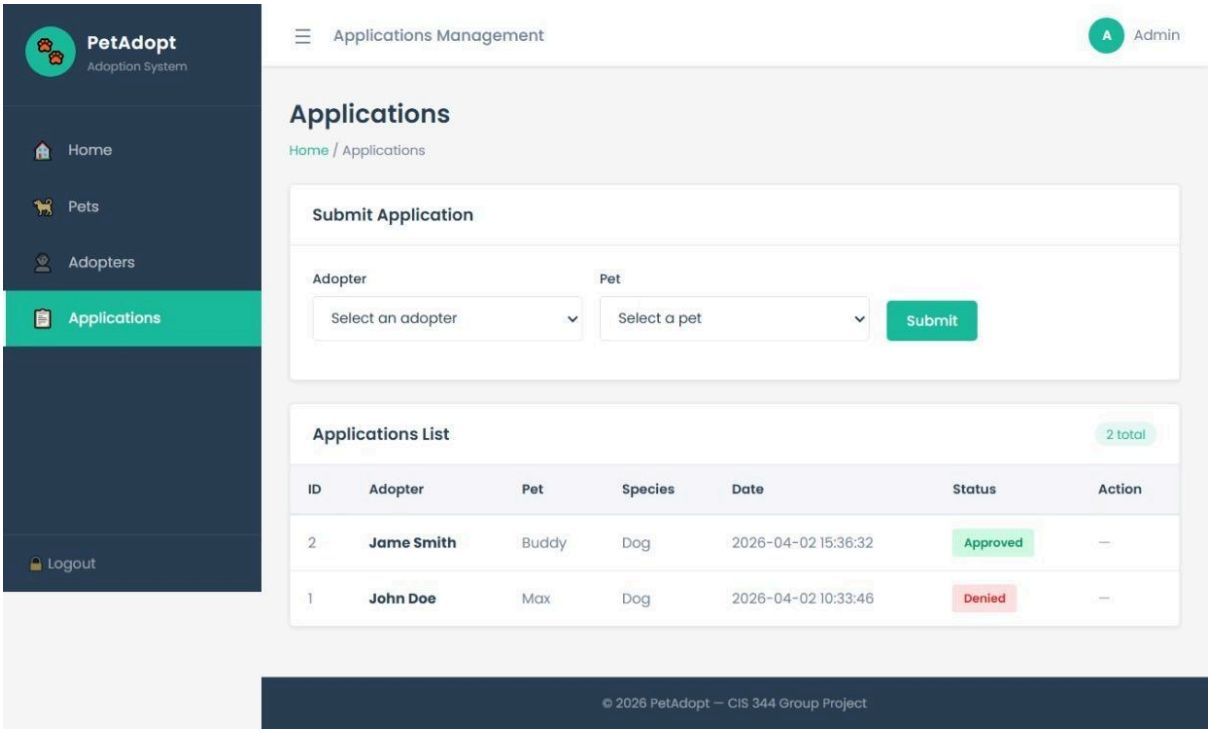

## Pets Page Screenshot:



**Adopters Page Screenshot:**




**Applications Page Screenshot:**

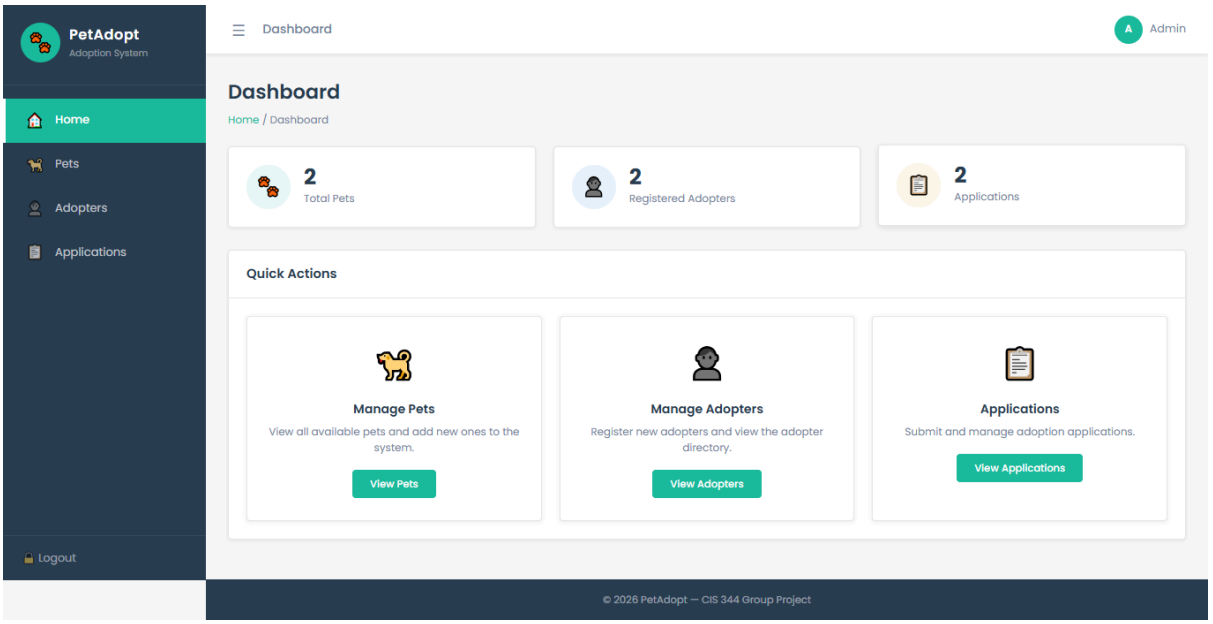


# 5. User Manual / Guide (Step-by-step)

## 5.1 Accessing the Application

1. Open http://localhost/CIS-344-GROUP-PROJECT in the browser.

2. It is opened on the application on the Dashboard/Homepage (no log-in).




## 5.2 Dashboard Overview

**2. Quick Actions:** At the bottom of the stats, one can find Quick Actions:

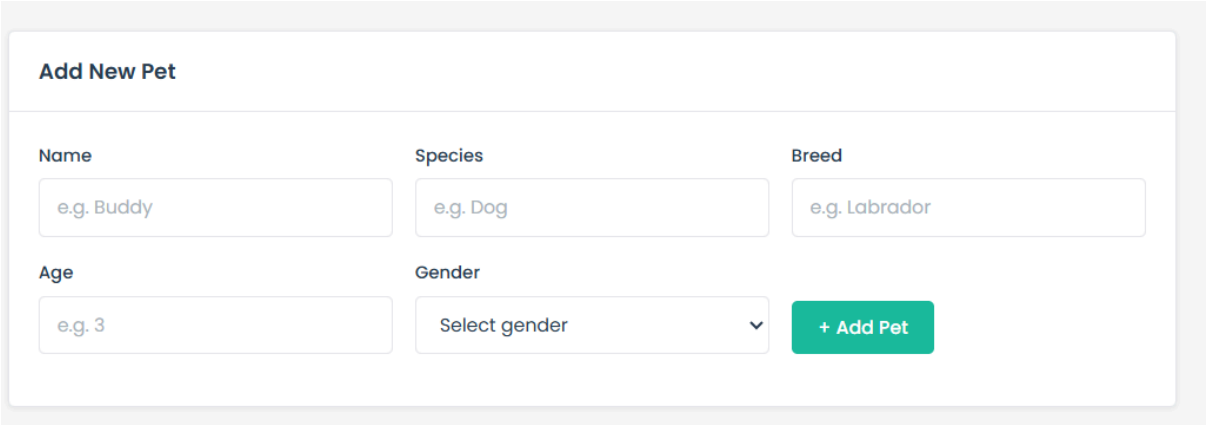
- Manage Pets: Transports you to Pets management page.
- Manage adopters: this brings one to the adopters page.

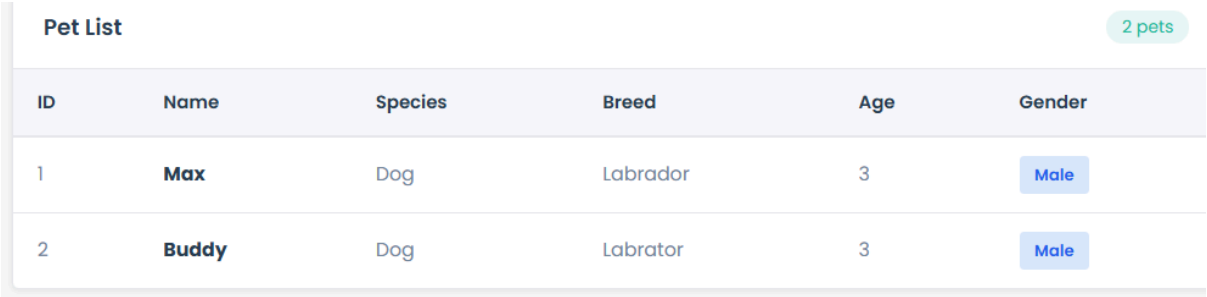
3. It have sidebar links that may be used to navigate to any page.

## 5.3 Managing Pets

1. Adding a New Pet: Click Add New Pet, Name, Species, Breed, Age, Gender Click Submit.

2. Pets List below presents the list of the current pets.



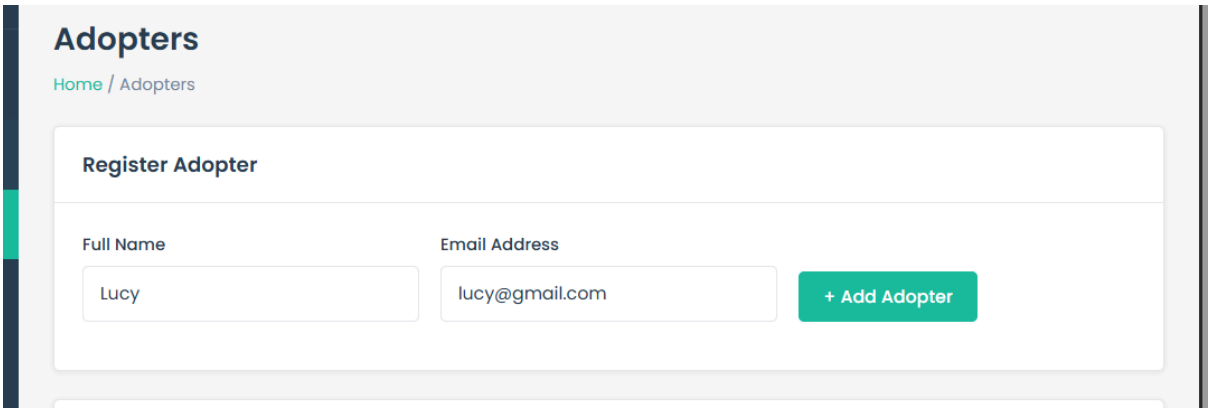



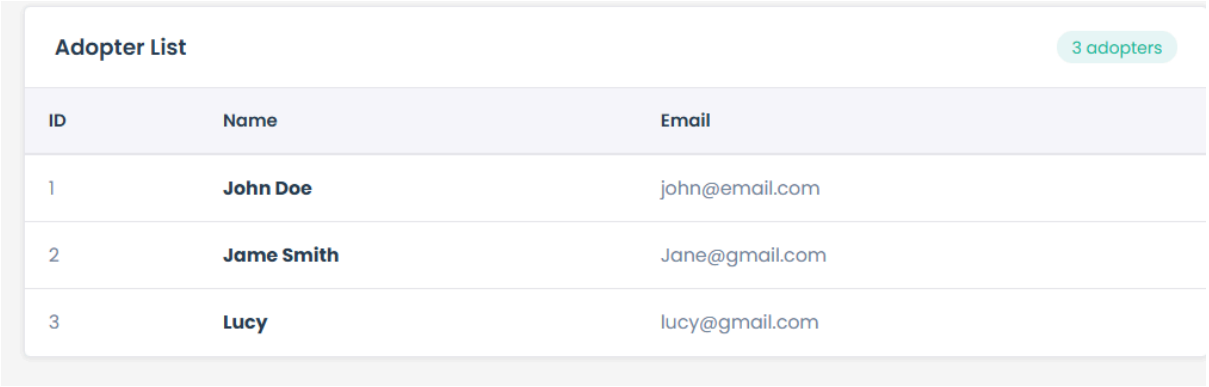

## 5.4 Managing Adopters

1.Click on the sidebar Adopters (or Manage Adopters on Dashboard)

2. Adding a New Adopter: Click on Add Adopter, Name and Email fill in and Click Submit.

3. Adopters List below This list contains all registered adopters.





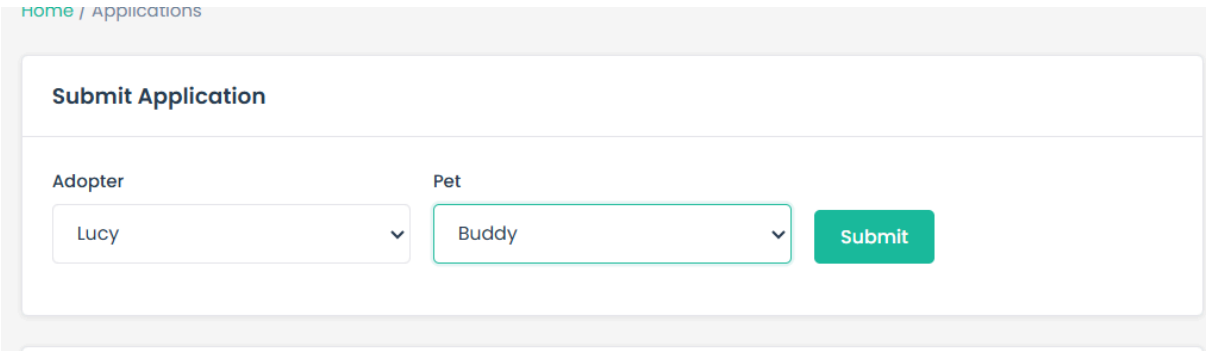

## 5.5 Managing Applications

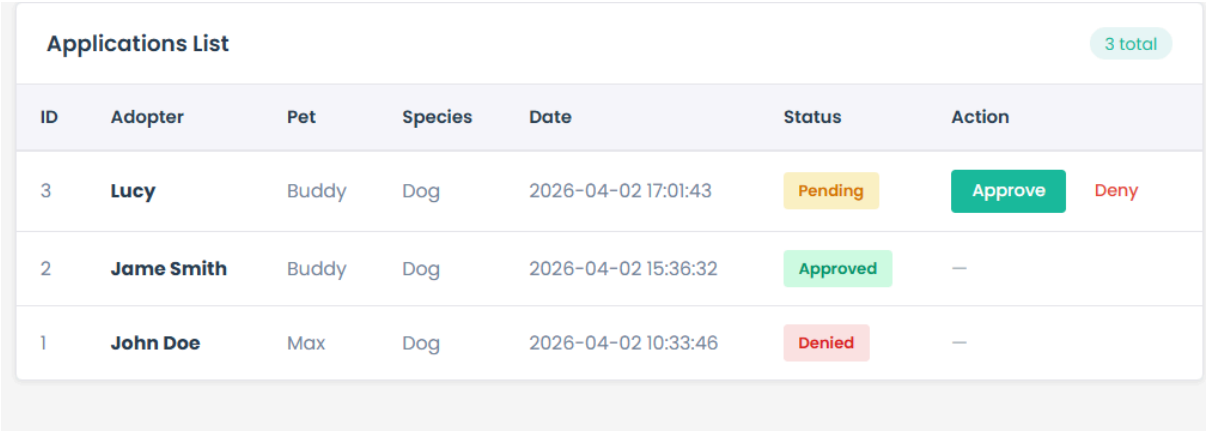
1. You should click on Applications on the side bar (or Applications on Dashboard).

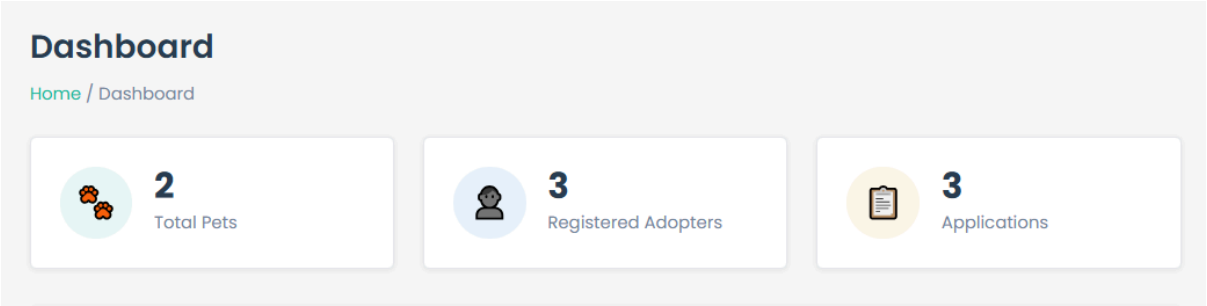
2. Submission of a New Application: Click Adopter, and then Click Pet, and then click Submit.

3. Applications List below represents all applications received.

4. Accept or Reject an Application: Click Accept or Reject next to the application Status changes instantly.






### 5.6 Notes & Tips

- Important: XAMPP Apache and MySQL should be open before accessing the system.
- The entire information is stored in MySQL database.
- Navigate the modules with ease with the navbar.